

ABHIGYAN DUTTA

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Education

PES University

Bengaluru, India

B.Tech in Computer Science & Engineering

2024 – Present

- **Relevant Coursework:** Operating Systems, Linux Administration, Computer Networks, Data Structures & Algorithms, Database Management, Cloud Computing, Software Engineering, Information Security.

Experience

OpenWISP

March 2026 – Present

Open Source Contributor

Remote

- Engineered a cross-repository **RBAC** mixin across distributed Django modules, resolving a critical cascade deletion vulnerability via URL-routing-based authorization; secured **ReadOnlyAdmin** endpoints against unauthorized mutations.
- Resolved a **5-year-old architectural limitation** in **openwisp-radius** by building non-strict CIDR and subnet validation, eliminating **AttributeError** crashes in a production system serving thousands of network devices.
- Architected a **concurrent, cache-safe** timezone pipeline (**zoneinfo**, **lru_cache**) preventing InfluxDB 500 errors; fixed a fatal **double free** CI memory corruption bug; authored **Pytest** suites maintaining **97%+ CI coverage**.

Projects

Self-Hosted Network & Media Controller | *Linux, systemd, Networking*

- Deployed a network-wide **DNS Sinkhole** filtering telemetry at the protocol level, reducing client bandwidth load by **15%**.
- Built a real-time transcoding pipeline with **Intel Iris Xe iGPU** hardware acceleration; tuned process scheduling to cut CPU load by **40%**.
- Hardened system security with **SSH Keys** and **ZeroTier** overlay networking, eliminating **100%** of public-facing port exposure.

PESU Content Automation Suite | *Python, Playwright, PyInstaller*

- Built a headless browser automation pipeline to scrape and archive course materials, cutting manual archival time by **95%**.
- Designed a **Regex-driven** file pipeline to parse and convert 100+ proprietary PPT formats into PDFs error-free across malformed inputs.
- Packaged cross-platform executables (.exe, .dmg, .deb) via PyInstaller for non-technical users across all major operating systems.

SemWork (Academic Resource Planner) | *React, Firebase, AI Integration*

- Built a data pipeline converting raw syllabus text into **NoSQL-compatible** JSON structures, reducing manual tracking time by **50%**.
- Designed an **idempotent sync protocol** resolving concurrent data conflicts across clients with **zero-loss** user state preservation.
- Integrated an AI Study Assistant with **sub-200ms** response latency using prompt engineering for context-aware study support.

Technical Skills

Languages: C,C++, Python, JavaScript, SQL, Bash

Systems & Infrastructure: Linux (Debian/Ubuntu), systemd, SSH, ZeroTier, TCP/IP, DNS

Tools: Git, Pytest, Playwright, REST APIs, CI/CD, Vercel/Netlify

Concepts: Data Structures & Algorithms, Distributed Systems, System Architecture, Cloud Infrastructure (AWS/GCP), Concurrency & Caching, Software Design

Soft Skills: Technical Documentation, Cross-functional Leadership, Agile Methodology, Problem Solving

Leadership Roles

Technical Lead, Vegavath | *Full-Stack Dev & Mentorship*

Nov 2025 – Present

- Built and maintained club platform; architected **CI/CD pipelines** achieving **40% faster deploys** and mentored a 10-person junior team.

Technical Lead, Ignition 1.0 Hackathon | *Cloud Infra & Event Management*

Nov 2025

- Orchestrated cloud infrastructure for 150+ concurrent users; provided live deployment mentorship with **zero downtime**.